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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the stage 3 protocol and data model for the Nimsas Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the IMS AS.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2] and TS 23.502 [3].

The IP Multimedia Subsystem (IMS) supporting DC architecture and procedures are specified in TS 23.228 [14].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[7] 3GPP TR 21.900: "Technical Specification Group working methods".

[8] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[9] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[10] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[11] IETF RFC 7540: "Hypertext Transfer Protocol Version 2 (HTTP/2)".

[12] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[13] IETF RFC 7807: "Problem Details for HTTP APIs".

[14] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".

[15] 3GPP TS 29.562: "5G System; Home Subscriber Server (HSS) Services; Stage 3".

[16] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

[17] 3GPP TS 29.176: "IP Multimedia Subsystems (IMS); Media Function (MF) Services; Stage 3".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.228 [14] apply:

- **Bootstrap data channel**
- **DC Application Server**

3.2 Symbols

For the purposes of the present document, the following symbols given in 3GPP TS 23.228 [14] apply:

Cr	Reference Point between an AS and an MRFC for media control.
Mr′	Reference Point between an AS and an MRFC for session control.
DC1	Reference Point between an SBI capable IMS AS and DCSF.
DC2	Reference Point between an SBI capable IMS AS and MF.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

AR	Augmented Reality
DC	Data Channel
DCSF	Data Channel Signalling Function
MF	Media Function

4 Overview

The IP Multimedia Subsystem (IMS) architecture supporting the Data Channel services is defined in 3GPP TS 23.228 [14].

The IP Multimedia Subsystem (IMS) Application Server (AS) is enhanced to offer services to DCSF via the Nimsas service-based interface (see 3GPP TS 23.228 [14]) and support the following functionalities:

- The IMS AS interacts with the DCSF via DC1 for event notifications;
- The IMS AS receives the data channel control instructions from the DCSF and accordingly interacts with the MF via DC2 or with MRF via Mr'/Cr for data channel media resource management.

Figure 4-1 provides the reference model (in service-based interface representation and in reference point representation), with focus on the IMS AS.

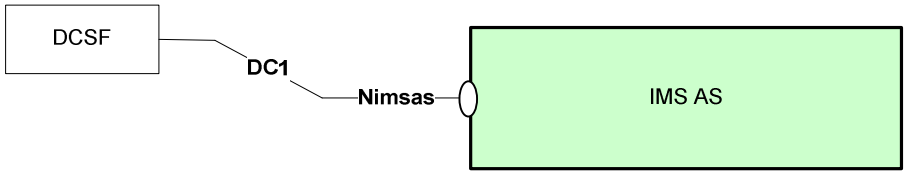

Figure 4-1: Reference model – IMS AS

DC1 is the reference point between an SBI capable IMS AS and DCSF.

5 Services offered by the IMS AS

5.1 Introduction

The IMS AS offers to other NFs the following services:

- Nimsas_SessionEventControl. This service enables the consumer to be notified about session events when served IMS subscribers take part in IMS sessions.
- Nimsas_MediaControl. This service enables the consumer to control IMS AS handling of media flow within an IMS session.

Table 5.1-1 summarizes the SBI services produced by the IMS AS and the corresponding APIs defined for this specification.

Table 5.1-1: API Descriptions

Service Name	Clause	Description	OpenAPI Specification File	apiName	Annex
Nimsas_SessionEventControl	6.1	IMS AS SessionEventControl Service	TS29175_Nimsas_SessionEventControl.yaml	nimsas-sec	A.2
Nimsas_MediaControl	6.2	IMS AS MediaControl Service	TS29175_Nimsas_MediaControl.yaml	nimsas-mc	A.3

5.2 Nimsas_SessionEventControl Service

5.2.1 Service Description

The Nimsas_SessionEventControl service enables the consumer to be notified about session events when served IMS subscribers take part in IMS sessions.

5.2.2 Service Operations

5.2.2.1 Introduction

The Service operations defined for the Nimsas_SessionEventControl service are as follows:

- Notify: It allows the IMS AS to notify the IMS session control events of a specific IMS subscriber to NFs. This service operation reports the event to the consumer that has subscribed implicitly.

NOTE: Explicit subscription to receive session events is another service operation defined in TS 23.228 [14] which has not specified in this Release. In this Release an implicit subscription is assumed where the IMS AS notifies a configured or discovered the NF consumer of a call event through a Notify.

Table 5.2.2.1-1: Service operations supported by the Nimsas_SessionEventControl service

Service Operations	Description	Operation Semantics	Example Consumer(s)
Notify	Notification about IMS session control events of a specific IMS subscriber.	Subscribe/Notify	DCSF

5.2.2.1A Subscribe

This is a pseudo operation, the DCSF does not actually provide Subscribe service operation through Nimsas_SessionEventControl service in this Release. The actual subscription is implicitly subscribed with the SessionEventNotificationUri.

5.2.2.2 Notify

5.2.2.2.1 General

This service operation is invoked by IMS AS and used to enables IMS AS to notify consumers of session events related to a specific served IMS subscriber requesting use of IMS media (e.g. data channel media).

5.2.2.2.2 Notification for Session Event

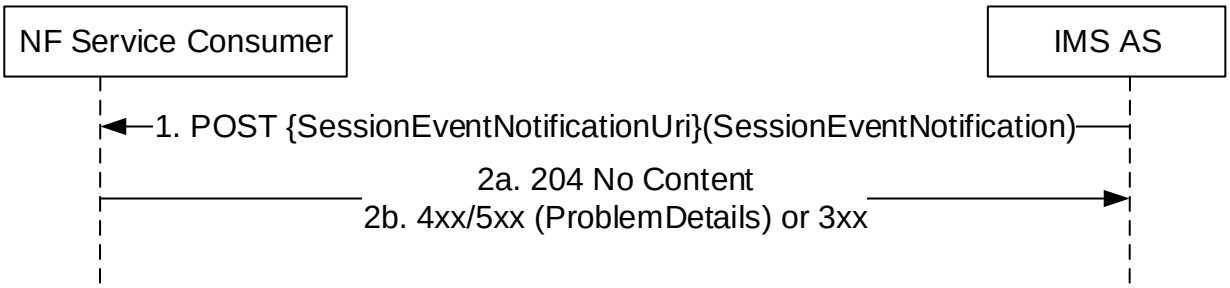

Figure 5.2.2.2.2-1: Notification for Session Event

1. If the IMS AS observes the IMS session control events of a specific IMS subscriber that requesting use of IMS data channel media, for which an NF service consumer has subscribed as specified in clause 5.2.2.2A, the IMS AS shall send a POST request to the SessionEventNotificationUri as specified in clause 6.1.5.2.2.

2a. Upon success, the NF Service Consumer responds with "204 No Content". The NF service consumer may invoke the Nimsas_MediaControl service before sending the session event notification response to IMS AS if the NF service consumer determines that media resources for data channel are needed.

2b. On failure or redirection:

- If the NF Service Consumer does not consider the "SessionEventNotificationUri" as a valid notification URI, the NF Service Consumer shall return "404 Not Found" status code with the ProblemDetails IE providing details of the error.

- In the case of redirection, the NF service consumer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service consumer endpoint.

5.3 Nimsas_MediaControl Service

5.3.1 Service Description

The Nimsas_MediaControl service enables the consumer to control IMS AS handling of media flow within an IMS session. The service can be used by the consumer before responding to a Nimsas_SessionEventControl_Notity request.

5.3.2 Service Operations

5.3.2.1 Introduction

The Service operations defined for the Nimsas_MediaControl service are as follows:

- MediaInstruction: It provides instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event, and that may require resource reservation in media resource. E.g., MF or MRF.

NOTE: For the case when a specific media flow needs to be terminated in MF media resource (i.e. termination of a Data Channel media descriptor offered by one of the IMS subscribers) or originated by the MF or MRF (i.e. origination of a data channel media descriptor to be offered towards one of the IMS subscribers), the consumer must provide a complete MF media specification including information required by MF to know how to terminate or originate the media flow.

Table 5.3.2.1-1: Service operations supported by the Nimsas_MediaControl service

Service Operations	Description	Operation Semantics	Example Consumer(s)
MediaInstruction	Instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event	Request/Response	DCSF

5.3.2.2 MediaInstruction

5.3.2.2.1 General

This service operation provides instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event, and that may require resource reservation in media resource

The MediaInstruction service operation defined for Nimsas_MediaControl service is used:

- to enable the NF consumer to instruct the IMS AS how to manage the MF media resource. E.g., setting up the bootstrap data channel with MF both for originating and terminating side, terminating the media flow of the originating UE to MF and updating the MF resource with MDC2 media endpoint information of DC Application Server.

5.3.2.2.2 Media Instruction

Figure 5.3.2.2.2-1 shows a scenario where the NF service consumer (e.g. DCSF) sends a request to the IMS AS to instruct the IMS AS how to manage the MF media resource. The request contains the IMS session identity (/{sessionId}) which is assigned by the IMS AS during the IMS session setup and will be notified to consumer via Nimsas_SessionEventControl notify service operation. After that, the consumer shall reuse the session ID it received from the IMS AS for referencing the same session.

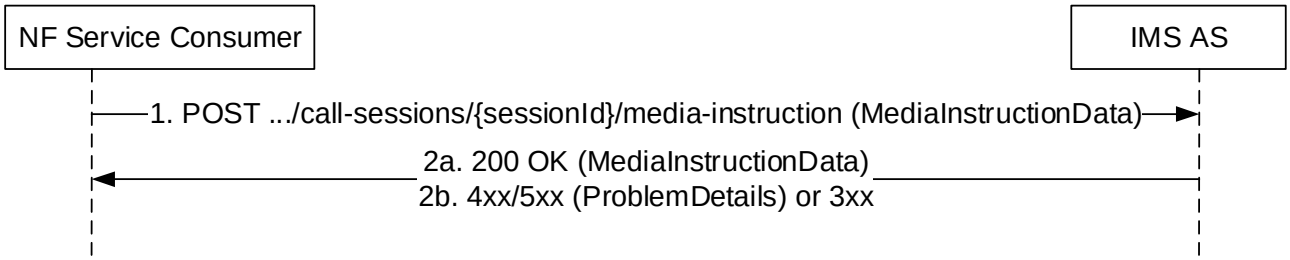

Figure 5.3.2.2.2-1: Media instruction from NF service consumer

1. The NF service consumer shall send a POST request to the resource representing the individual session resource in the IMS AS. The payload body of the POST request shall contain the media instructions for all the media

flows that the NF Service Consumer want to control based on its policies for the received IMS session event via Nimsas_SessionEventControl notification.

2a. Upon success, the IMS AS responds with "200 OK" with the message body containing a representation of the created media instruction data.

2b. On failure or redirection:

In the case of failure, one of the HTTP status code listed in Table 6.2.3.2.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.2.3.2.4.2.2-2.

In the case of redirection, the NF service producer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service producer endpoint.

6 API Definitions

6.1 Nimsas_SessionEventControl Service API

6.1.1 API URI

The Nimsas_SessionEventControl service shall use the Nimsas_SessionEventControl API.

The API URI of the Nimsas_SessionEventControl API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "nimsas-sec".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

6.1.2 Usage of HTTP

6.1.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_SessionEventControl API is contained in Annex A.

6.1.2.2 HTTP standard headers

6.1.2.2.1 General

See clause 5.2.2 of 3GPP TS 29.500 [4] for the usage of HTTP standard headers.

6.1.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of 3GPP TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

6.1.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of 3GPP TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of 3GPP TS 29.500 [4] may be supported.

6.1.3 Resources

6.1.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.1.3.1-1 depicts the resource URIs structure for the Nimsas_SessionEventControl service API.

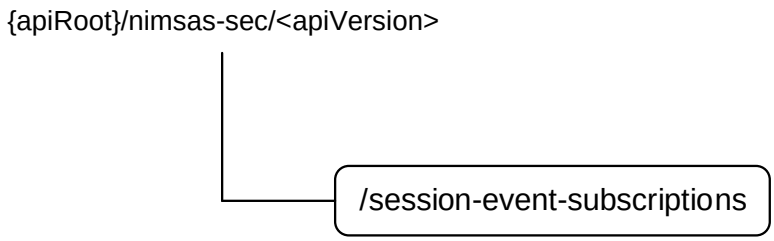

Figure 6.1.3.1-1: Resource URI structure of the Nimsas_SessionEventControl API

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.1.3.1-1: Resources and methods overview

Resource purpose/name	Resource URI (relative path after API URI)	HTTP method or custom operation	Description (service operation)
Session Event Subscriptions (Collection)	/session-event-subscriptions	POST	This is a pseudo resource.

6.1.3.2 Resource: Session Event Subscriptions

6.1.3.2.1 Description

6.1.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-sec/<apiVersion>/session-event-subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

Table 6.1.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

6.1.3.2.3 Resource Standard Methods

6.1.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.1.3.2.3.1-1.

Table 6.1.3.2.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description	Applicability
n/a					

This method shall support the request data structures specified in table 6.1.3.2.3.1-2 and the response data structures and response codes specified in table 6.1.3.2.3.1-3.

Table 6.1.3.2.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
Any			

Table 6.1.3.2.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a				
NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.1.3.2.4 Resource Custom Operations

None.

6.1.4 Custom Operations without associated resources

None in this release of the specification.

6.1.5 Notifications

6.1.5.1 General

Notifications shall comply to clause 6.2 of 3GPP TS 29.500 [4] and clause 4.6.2.3 of 3GPP TS 29.501 [5].

Table 6.1.5.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
Session Event Notification	{SessionEventNotificationUri}	POST	Report the session event observed.

6.1.5.2 Session Event Notification

6.1.5.2.1 Description

The Session Event Notification is used by the IMS AS to report the observed Session Event to notification endpoints.

6.1.5.2.2 Target URI

The Callback URI "{SessionEventNotificationUri}" shall be used with the callback URI variables defined in table 6.1.5.2.2-1.

Table 6.1.5.2.2-1: Callback URI variables

Name	Definition
SessionEventNotificationUri	The notification URI of session event Consumer where to receive the requested session event. The SessionEventNotificationUri is not provided by NF service consumer via Nimsas_SessionEventControl API in this Release.

6.1.5.2.3 Standard Methods

6.1.5.2.3.1 POST

This method shall support the request data structures specified in table 6.1.5.2.3.1-1 and the response data structures and response codes specified in table 6.1.5.2.3.1-1.

Table 6.1.5.2.3.1-1: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
SessionEventNotification	M	1	The session event notification to notification endpoints.

Table 6.1.5.2.3.1-2: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	This case represents a successful notification of the event.
ProblemDetails	O	0..1	404 Not Found	If the NF Service Consumer considers the "SessionEventNotificationUri" is not recognized, the NF Service Consumer shall return "404 Not Found" status code
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].				

Table 6.1.5.2.3-3: Headers supported by the 307 Response Code on this endpoint

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF instance ID towards which the notification is redirected

Table 6.1.5.2.3-4: Headers supported by the 308 Response Code on this endpoint

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF instance ID towards which the notification is redirected

6.1.6 Data Model

6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Nimsas_SessionEventControl service based interface protocol.

Table 6.1.6.1-1: Nimsas_SessionEventControl specific Data Types

Data type	Clause defined	Description
SessionEventNotification	6.1.6.2.2	The session event notification item.
NotificationEvent	6.1.6.2.3	The notification event information.
SessionInfo	6.1.6.2.4	The session information related to the notification event.
MediaInfo	6.1.6.2.5	The media information related to the notification event.
MediaSpecification	6.1.6.2.6	The media specification information which includes media description attributes.
EventType	6.1.6.3.3	The type of the event for which the notification is generated.
MediaType	6.1.6.3.4	The type of the media for which the notification is generated.
SessionCase	6.1.6.3.5	The type of the session case.
EventInitiator	6.1.6.3.6	The initiator of the event for which the notification is generated.

Table 6.1.6.1-2 specifies data types re-used by the Nimsas_SessionEventControl service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_SessionEventControl service based interface.

Table 6.1.6.1-2: Nimsas_SessionEventControl re-used Data Types

Data type	Reference	Comments
ImsPublicId	3GPP TS 29.562 [15]	IMS Public Identity.
DcStream	3GPP TS 29.571 [16]	Data Channel mapping and configuration information.
MaxMessageSize	3GPP TS 29.571 [16]	Maximum message size of to be expected.
DcEndpoint	3GPP TS 29.571 [16]	local or remote endpoint for the Data Channel.
SecuritySetup	3GPP TS 29.571 [16]	Security setup of the DTLS connection.
SessionId	3GPP TS 29.571 [16]	IMS Session Identity.
MediaId	3GPP TS 29.571 [16]	IMS media flow identity.

6.1.6.2 Structured data types

6.1.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.1.6.2.2 Type: SessionEventNotification

Table 6.1.6.2.2-1: Definition of type SessionEventNotification

Attribute name	Data type	P	Cardinality	Description
notificationEvent	NotificationEvent	M	1	The NotificationEvent corresponds to a report on one subscribed event.
sessionId	SessionId	M	1	The session ID is the identity of the IMS session for which the event relates to. This IE contains the information in the Call-ID header which is the typical header of SIP message.
sessionInfo	SessionInfo	C	0..1	The sessionInfo is the IMS session related information, e.g. the calling and called identity.
mediaInfoList	map(MediaInfo)	C	1..N	Media info list includes a list of media related information, e.g. media identity, media type and media specifications. The mediaId attribute within the MediaInfo data type is the key of the map.

6.1.6.2.3 Type: NotificationEvent

Table 6.1.6.2.3-1: Definition of type NotificationEvent

Attribute name	Data type	P	Cardinality	Description
eventType	EventType	M	1	The event type of the event for which the notification is generated.
eventInitiator	EventInitiator	O	0..1	Event initiator indicates initiator of the event, e.g. "served IMS subscriber" vs "remote IMS subscriber".

6.1.6.2.4 Type: SessionInfo

Table 6.1.6.2.4-1: Definition of type SessionInfo

Attribute name	Data type	P	Cardinality	Description
callingIdentity	ImsPublicId	O	0..1	The public identity of the calling IMS subscriber.
calledIdentity	ImsPublicId	O	0..1	The public identity of the called IMS Subscriber.
sessionCase	SessionCase	O	0..1	Indicator to indicate that whether this is an originating or terminating IMS session.

6.1.6.2.5 Type: MediaInfo

Table 6.1.6.2.5-1: Definition of type MediaInfo

Attribute name	Data type	P	Cardinality	Description
mediaId	MediaId	M	1	The media ID uniquely identifies this media item within the list. The identity is allocated by IMS AS.
mediaType	MediaType	M	1	The media type of the media for which the notification is generated.
dcMediaSpecification	DcMediaSpecification	C	0..1	The data channel media specification includes the relevant media attributes of interest to the consumer. It shall be contained if the mediaType is set to "DC".

6.1.6.2.6 Type: DcMediaSpecification

Table 6.1.6.2.6-1: Definition of type DcMediaSpecification

Attribute name	Data type	P	Cardinality	Description
streams	map(DcStream)	M	1..N	Represents the data channel mapping and configuration information when originating/terminating data channel media flows on the Mb interface. The streamId attribute within the DcStream data type is the key of the map. (NOTE)
maxMessageSize	MaxMessageSize	C	0..1	Represents the maximum message size of to be expected. (NOTE)
localDcEndpoint	DcEndpoint	C	0..1	Represents the local Data channel endpoint. (NOTE)
remoteDcEndpoint	DcEndpoint	C	0..1	Represents the remote port for the Data Channel. (NOTE)
securitySetup	SecuritySetup	C	0..1	Represents the security set up of the DTLS association. (NOTE)
NOTE: The attribute and the value are derived from the SDP received by the IMS AS in a SIP INVITE or a re-INVITE related to an IMS Data Channel and the corresponding DTLS connection.				

6.1.6.3 Simple data types and enumerations

6.1.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.1.6.3.2 Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.

Table 6.1.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability

6.1.6.3.3 Enumeration: EventType

The enumeration EventType represents the type of the event for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.3-1.

Table 6.1.6.3.3-1: Enumeration EventType

Enumeration value	Description
"SESSION_ESTABLISHMENT_REQUEST"	Session establishment request event.
"SESSION_ESTABLISHMENT_PROGRESS"	Session establishment progress event.
"SESSION_ESTABLISHMENT_ALERTING"	Session establishment alerting event.
"SESSION_ESTABLISHMENT_SUCCESS"	Session establishment success event.
"SESSION_ESTABLISHMENT_FAILURE"	Session establishment failure event.
"MEDIA_CHANGE_REQUEST"	Media change request event.
"MEDIA_CHANGE_SUCCESS"	Media change success event.
"MEDIA_CHANGE_FAILURE"	Media change failure event.
"SESSION_TERMINATION"	Session termination event.

6.1.6.3.4 Enumeration: MediaType

The enumeration MediaType represents the type of the media for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.4-1.

Table 6.1.6.3.4-1: Enumeration MediaType

Enumeration value	Description
"DC"	Data channel media type.
"AUDIO"	Audio media type.
"VIDEO"	Video media type.

6.1.6.3.5 Enumeration: SessionCase

The enumeration SessionCase represents the type of the session to for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.5-1.

Table 6.1.6.3.5-1: Enumeration SessionCase

Enumeration value	Description
"ORIGINATING_IMS_SESSION"	Originating IMS session.
"TERMINATING_IMS_SESSION"	Terminating IMS session.

6.1.6.3.6 Enumeration: EventInitiator

The enumeration eventInitiator represents the initiator of the event for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.6-1.

Table 6.1.6.3.6-1: Enumeration EventInitiator

Enumeration value	Description
"SERVED_IMS_SUBSCRIBER"	Served IMS subscriber.
"REMOTE_IMS_SUBSCRIBER"	Remote IMS subscriber.

6.1.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

6.1.6.5 Binary data

None in this release of the specification.

6.1.7 Error Handling

6.1.7.1 General

For the Nimsas_SessionEventControl API, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_SessionEventControl API.

6.1.7.2 Protocol Errors

No specific procedures for the Nimsas_SessionEventControl service are specified.

6.1.7.3 Application Errors

The application errors defined for the Nimsas_SessionEventControl service are listed in Table 6.1.7.3-1.

Table 6.1.7.3-1: Application errors

Application Error	HTTP status code	Description

6.1.8 Feature negotiation

The optional features in table 6.1.8-1 are defined for the Nimsas_SessionEventControl API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

Table 6.1.8-1: Supported Features

Feature number	Feature Name	Description

6.1.9 Security

As indicated in 3GPP TS 33.501 [8] and 3GPP TS 29.500 [4], the access to the Nimsas_SessionEventControl API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_SessionEventControl API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_SessionEventControl service.

The Nimsas_SessionEventControl API defines a single scope "nimsas-sessioneventcontrol" for the entire service, and it does not define any additional scopes at resource or operation level.

6.1.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see 3GPP TS 29.500 [4]).

An SCP that reselects a different NF consumer (e.g. DCSF) instance will return the NF Instance ID of the new NF consumer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of 3GPP TS 29.500 [4].

If an NF consumer (e.g. DCSF) redirects a service request to a different NF consumer using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new NF consumer towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of 3GPP TS 29.500 [4].

6.2 Nimsas_MediaControl Service API

6.2.1 API URI

The Nimsas_MediaControl service shall use the Nimsas_MediaControl API.

The API URI of the Nimsas_MediaControl API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "nimsas-mc".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

6.2.2 Usage of HTTP

6.2.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_MediaControl API is contained in Annex A.

6.2.2.2 HTTP standard headers

6.2.2.2.1 General

See clause 5.2.2 of 3GPP TS 29.500 [4] for the usage of HTTP standard headers.

6.2.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of 3GPP TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

6.2.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of 3GPP TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of 3GPP TS 29.500 [4] may be supported.

6.2.3 Resources

6.2.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.2.3.1-1 depicts the resource URIs structure for the Nimsas_MediaControl service API.

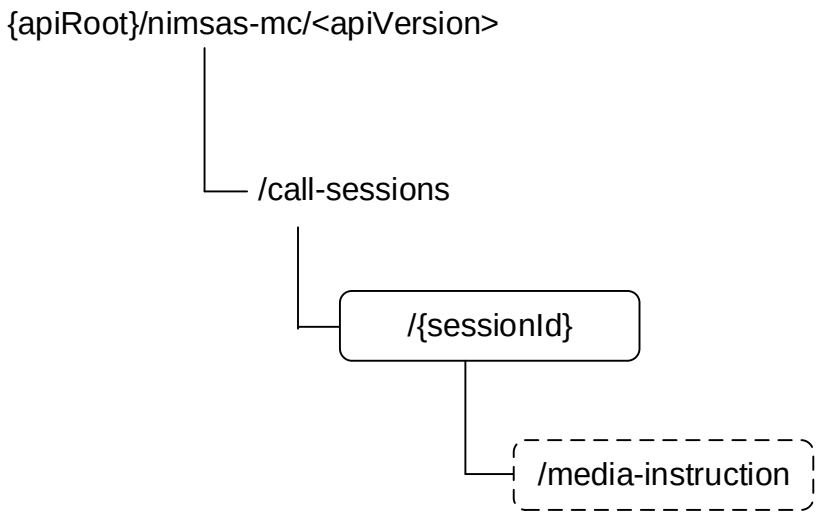

Figure 6.2.3.1-1: Resource URI structure of the Nimsas_MediaControl API

Table 6.2.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.2.3.1-1: Resources and methods overview

Resource purpose/name	Resource URI (relative path after API URI)	HTTP method or custom operation	Description (service operation)
Individual call session (Document)	/call-sessions/{sessionId}/media-instruction	media-instruction (POST)	Update

6.2.3.2 Resource: Individual call session

6.2.3.2.1 Description

This resource represents an individual call session created in the IMS AS.

This resource is modelled with the Document resource archetype (see clause C.1 of 3GPP TS 29.501 [5]).

6.2.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-mc/<apiVersion>/call-sessions/{sessionId}**

This resource shall support the resource URI variables defined in table 6.2.3.2.2-1.

Table 6.2.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.2.1
sessionId	SessionId	Session ID assigned by the IMS AS during the IMS session setup and will be notified to consumer via Nimsas_SessionEventControl service. The consumer shall reuse the session ID it received from the IMS AS for referencing the same session.

6.2.3.2.3 Resource Standard Methods

None.

6.2.3.2.4 Resource Custom Operations

6.2.3.2.4.1 Overview

Table 6.2.3.2.4.1-1: Custom operations

Operation Name	Custom operation URI	Mapped HTTP method	Description
media-instruction	{resourceUri}/media-instruction	POST	Update call session service operation.

6.2.3.2.4.2 Operation: media-instruction

6.2.3.2.4.2.1 Description

6.2.3.2.4.2.2 Operation Definition

This custom operation updates an individual call session resource and provide a set of media instruction information for a given IMS session, towards IMS AS.

This operation shall support the request data structures specified in table 6.2.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.2.3.2.4.2.2-2.

Table 6.2.3.2.4.2.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
MediaInstructionData	M	1	Representation of the updates to apply to the call session.

Table 6.2.3.2.4.2.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
MediaInstructionData	C	0..1	200 OK	Successful update of the call session, when the IMS AS needs to return information in the response.
n/a			204 No Content	Successful update of the call session, when the IMS AS does not need to return information in the response.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	400 Bad Request	Indicates the media instruction has failed due to application error. The "cause" attribute may be used to indicate one of the following application errors: - MEDIA_ID_NOT_FOUND
NOTE 1: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].				

Table 6.2.3.3.4.2.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF instance ID towards which the request is redirected.

Table 6.2.3.3.4.2.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target SMF (service) instance ID towards which the request is redirected.

6.2.4 Custom Operations without associated resources

None in this release of the specification.

6.2.5 Notifications

None in this release of the specification.

6.2.6 Data Model

6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Nimsas_MediaControl service based interface protocol.

Table 6.2.6.1-1: Nimsas_MediaControl specific Data Types

Data type	Clause defined	Description
ArMediaSpecification	6.2.6.2.5	The AR media specification information needed for AR communication services from application layer.
DcMediaSpecification	6.2.6.2.4	The data channel media specification information needed for data channel media stream from application layer.
MediaInstruction	6.2.6.3.3	The media instruction for handling a specific media flow.
MediaInstructionData	6.2.6.2.2	The media instruction data for a specific session.
MediaInstructions	6.2.6.2.3	The media instructions for a specific media flow.

Table 6.2.6.1-2 specifies data types re-used by the Nimsas_MediaControl service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_MediaControl service based interface.

Table 6.2.6.1-2: Nimsas_MediaControl re-used Data Types

Data type	Reference	Comments
DcStream	3GPP TS 29.571 [16]	Data Channel mapping and configuration information.
Endpoint	3GPP TS 29.571 [16]	IP Endpoint information used for MDC1 and MDC2.
MaxMessageSize	3GPP TS 29.571 [16]	Maximum SCTP user message size.
Mdc2Protocol	3GPP TS 29.176 [17]	Transport layer protocols for MDC2 interface.
MediaId	3GPP TS 29.571 [16]	IMS media flow identity.
MediaProxy	3GPP TS 29.571 [16]	Media proxy configuration applicable to the media flow.
MediaResourceType	3GPP TS 29.571 [16]	IMS media resource type.
ReplaceHttpUrl	3GPP TS 29.571 [16]	A list of replacement HTTP URL per stream ID.
SessionId	3GPP TS 29.571 [16]	IMS session identity.
Uri	3GPP TS 29.571 [16]	URI

6.2.6.2 Structured data types

6.2.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.2.6.2.2 Type: MediaInstructionData

Table 6.2.6.2.2-1: Definition of type MediaInstructionData

Attribute name	Data type	P	Cardinality	Description
sessionId	SessionId	M	1	The session ID is the identity of the IMS session for which the MediaInstructions applies.
mediaInstructionSet	map(MediaInstructions)	M	1..N	A map (list of key-value pairs where mediaId as key) of MediaInstructions. The mediaInstructionSet includes a set of instructions for each media flow to control.

6.2.6.2.3 Type: MediaInstructions

Table 6.2.6.2.3-1: Definition of type MediaInstructions

Attribute name	Data type	P	Cardinality	Description
mediaId	MediaId	M	1	The media ID uniquely identifies this media item within the list and used by the producer (IMS AS) and the consumer for referencing purposes. The consumer reuses the media ID it received from the IMS AS for referencing the same media. This field will be null for instructions related to originating new media.
mediaResourceType	MediaResourceType	M	1	The mediaResourceType identify the Media Resource types the Media instruction is intended for.
mediaInstruction	MediaInstruction	C	0..1	mediaInstruction includes instructions to the producer (IMS AS) for handling the media.
dcMediaSpecification	DcMediaSpecification	C	0..1	dcMediaSpecification includes the description of additional media specification information which are needed for data channel media stream from application layer. It shall be contained if the mediaResourceType is set to "DC".
arMediaSpecification	ArMediaSpecification	C	0..1	arMediaSpecification includes the description of additional media specification information which are needed for AR communication services from application layer. It shall be contained if the mediaResourceType is set to "AR".
mediaProcessingUrl	Uri	C	0..1	The mediaProcessingURL indicates the address where MF receive service-related media instructions. This field shall be contained in the response when originating a new media.

6.2.6.2.4 Type: DcMediaSpecification

Table 6.2.6.2.4-1: Definition of type DcMediaSpecification

Attribute name	Data type	P	Cardinality	Description
mediaProxyConfig	MediaProxy	C	0..1	Represents the media proxy configuration. It shall be included if the mediaId represents the bootstrap or P2A/A2P application data channel.
replaceHttpUrls	map(ReplaceHttpUrl)	C	1..N	Represents a list of replacement HTTP URL per stream ID allocated by the application layer for the specific IMS subscriber when requesting the application list (e.g. graphical user interface) via the MDC1 interface. It shall be included if the mediaId represents a bootstrap data channel and streamId is 0 or 100. The streamId attribute within the ReplaceHttpUrl data type is the key of the map.
mdc1EndpointDcsf	Endpoint	C	0..1	Represents the MDC1 media endpoint information reserved on DCSF, i.e. the IP address and port number of DCSF. It shall be included in the request of the media instruction if the mediaId represents a bootstrap data channel.
mdc1EndpointMf	Endpoint	C	0..1	Represents the negotiated MDC1 media endpoint information on MF, i.e. the IP address and port number of DC capable MF. It shall be included in the request of the media instruction if the mediaId represents a bootstrap data channel.
mdc2EndpointDcAs	Endpoint	C	0..1	Represents the negotiated MDC2 media endpoint information on DC AS, i.e. the IP address and port number of DC AS. It shall be included if in the request of the media instruction the mediaId represents a P2A/A2P application data channel and the mediaProxyConfig is set "HTTP".
mdc2EndpointMf	Endpoint	C	0..1	Represents the negotiated MDC2 media endpoint information on MF, i.e. the IP address and port number of DC capable MF. It shall be included in the request of the media instruction if the mediaId represents a P2A/A2P application data channel and the mediaProxyConfig is set "HTTP".
mdc2Protocol	Mdc2Protocol	O	0..1	Represents the transport layer protocols for MDC2 interface. It may be included when Mdc2EndpointDcas is set "HTTP" and remoteMdc2Endpoint is present.
streams	map(DcStream)	M	1..N	Represents the data channel mapping and configuration information when originating/terminating data channel media flows on the Mb interface. The streamId attribute within the DcStream data type is the key of the map.

6.2.6.2.5 Type: ArMediaSpecification

Table 6.2.6.2.5-1: Definition of type ArMediaSpecification

Attribute name	Data type	P	Cardinality	Description
mediaProcessingSpec	string	M	1	It specifies how the AR media stream should be processed. It indicates how the MF assists in the AR media rendering function.

6.2.6.3 Simple data types and enumerations

6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

Table 6.2.6.3.2-1: Simple data types

Type Name	Type Definition	Description

6.2.6.3.3 Enumeration: MediaInstruction

The enumeration MediaInstruction represents the instructions to the producer (IMS AS) for handling the media. It shall comply with the provisions defined in table 6.2.6.3.3-1.

Table 6.2.6.3.3-1: Enumeration MediaInstruction

Enumeration value	Description
"TERMINATE_MEDIA"	Terminate the offered media descriptor of the UE in the media resource.
"ORIGINATE_MEDIA"	Originate and offer a media descriptor from the mediaResource to the UE.
"TERMINATE_AND_ORIGINATE_MEDIA"	Terminate the offered media flow in the mediaResource from the UE and originate the same media flow from the mediaResource to the other UE.
"UPDATE_MEDIA"	Update a media flow of the mediaResource previously allocated by the instructions "TerminateMedia", "OriginateMedia" and "TerminateAndOriginateMedia".
"DELETE_MEDIA"	Delete a media flow of the mediaResource previously allocated by the instructions "TerminateMedia", "OriginateMedia" and "TerminateAndOriginateMedia".
"REJECT_MEDIA"	Remove an offered media flow.

6.2.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

6.2.6.5 Binary data

None in this release of the specification.

6.2.7 Error Handling

6.2.7.1 General

For the Nimsas_MediaControl API, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_MediaControl API.

6.2.7.2 Protocol Errors

No specific procedures for the Nimsas_MediaControl service are specified.

6.2.7.3 Application Errors

The application errors defined for the Nimsas_MediaControl service are listed in table 6.2.7.3-1.

Table 6.2.7.3-1: Application errors

Application Error	HTTP status code	Description
MEDIA_ID_NOT_FOUND	400 Bad Request	Indicates that the requested mediaId is not found in the IMS AS.

6.2.8 Feature negotiation

The optional features in table 6.2.8-1 are defined for the Nimsas_MediaControl API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

Table 6.2.8-1: Supported Features

Feature number	Feature Name	Description

6.2.9 Security

As indicated in 3GPP TS 33.501 [8] and 3GPP TS 29.500 [4], the access to the Nimsas_MediaControl API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_MediaControl API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_MediaControl service.

The Nimsas_MediaControl API defines a single scope " nimsas-mc " for the entire service, and it does not define any additional scopes at resource or operation level.

6.2.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see 3GPP TS 29.500 [4]).

An SCP that reselects a different IMS AS producer instance will return the NF Instance ID of the new IMS AS producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of 3GPP TS 29.500 [4].

If an IMS AS redirects a service request to a different IMS AS using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new IMS AS towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of 3GPP TS 29.500 [4].

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the API(s) defined in the present specification. It consists of OpenAPI specifications in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE 1: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5.3.1 of 3GPP TS 29.501 [5] and clause 5B of 3GPP TR 21.900 [7]).

A.2 Nimsas_SessionEventControl API

```
openapi: 3.0.0

info:

  title: 'IMS AS SessionEventControl Service'
  version: 1.0.0-alpha.2
  description: |
    Nimsas_SessionEventControl Service.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V18.0.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175/'

servers:
  - url: '{apiRoot}/nimsas-sec/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-sec

paths:
  /session-event-subscriptions:
    post:
    # This is a pseudo operation, NF service consumers shall NOT invoke this method!
      summary: Nimsas_SessionEventControl Subscribe service operation, pseudo operation
      tags:
        - Session Event Subscriptions (Collection)
      operationId: Subscribe
      requestBody:
        required: true
        content:
          application/json:
            # Unspecified schema for the JSON body, since this is not used by consumer or producer.
            schema: {}
      responses:
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
```

```
      callbacks:
        eeNotification:
          '{SessionEventNotificationUri}':
          # The URI in {SessionEventNotificationUri} is not provided by NF service consumer via
Nimsas_SessionEventControl API in this Release.
            post:
              requestBody:
                required: true
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/SessionEventNotification'
              responses:
                '204':
                  description: No Content, notification was succesfull.
                '307':
                  description: Temporary Redirect
                  content:
                    application/json:
                      schema:
                        $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                  headers:
                    Location:
                      description: >
                        The URI pointing to the resource located on the redirect target NF
                        service consumer.
                      required: true
                      schema:
                        type: string
                '308':
                  description: Permanent Redirect
                  content:
                    application/json:
                      schema:
                        $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                  headers:
                    Location:
                      description: >
                        The URI pointing to the resource located on the redirect target NF
                        service consumer.
                      required: true
                      schema:
                        type: string
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
                default:
                  $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
```

```
            nimsas-sec: Access to the Nimsas_SessionEventControl API

  schemas:

# STRUCTURED DATA TYPES

    SessionEventNotification:
      description: The session event notification item.
      type: object
      required:
        - notificationEvent
        - sessionId
      properties:
        notificationEvent:
          $ref: '#/components/schemas/NotificationEvent'
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        sessionInfo:
          $ref: '#/components/schemas/SessionInfo'
        mediaInfoList:
          description: >
            Contains a media related information. The mediaId attribute within the MediaInfo
            data type is the key of the map.
          type: object
          additionalProperties:
            $ref: '#/components/schemas/MediaInfo'
          minItems: 1

    NotificationEvent:
      description: The notification event information.
      type: object
      required:
        - eventType
      properties:
        eventType:
          $ref: '#/components/schemas/EventType'
        eventInitiator:
          $ref: '#/components/schemas/EventInitiator'

    SessionInfo:
      description: The session information related to the notification event.
      type: object
      properties:
        callingIdentity:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        calledIdentity:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        sessionCase:
          $ref: '#/components/schemas/SessionCase'

    MediaInfo:
      description: The media information related to the notification event.
      type: object
      required:
        - mediaId
        - mediaType
      properties:
        mediaId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaId'
        mediaType:
          $ref: '#/components/schemas/MediaType'
        dcMediaSpecification:
          $ref: '#/components/schemas/DcMediaSpecification'

    DcMediaSpecification:
      description: >
        The data channel media specification information contains media description attributes.
      type: object
      required:
        - streams
      properties:
        streams:
          description: >
            Contains a data channel mapping and configuration information. The streamId
            attribute within the DcStream data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/DcEndpoint'
```

```
          minItems: 1
        maxMessageSize:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MaxMessageSize'
        localDcEndpoint:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DcEndpoint'
        remoteDcEndpoint:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DcEndpoint'
        securitySetup:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SecuritySetup'

# ENUMS

    EventType:
      description: The type of the event for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - SESSION_ESTABLISHMENT_REQUEST
            - SESSION_ESTABLISHMENT_PROGRESS
            - SESSION_ESTABLISHMENT_ALERTING
            - SESSION_ESTABLISHMENT_SUCCESS
            - SESSION_ESTABLISHMENT_FAILURE
            - MEDIA_CHANGE_REQUEST
            - MEDIA_CHANGE_SUCCESS
            - MEDIA_CHANGE_FAILURE
            - SESSION_TERMINATION
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    MediaType:
      description: The type of the media for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - DC
            - AUDIO
            - VIDEO
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    SessionCase:
      description: The type of the session case.
      anyOf:
        - type: string
          enum:
            - ORIGINATING_IMS_SESSION
            - TERMINATING_IMS_SESSION
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    EventInitiator:
      description: The initiator of the event for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - SERVED_IMS_SUBSCRIBER
            - REMOTE_IMS_SUBSCRIBER
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.
```

A.3 Nimsas_MediaControl API

```
openapi: 3.0.0

info:
```

```
  title: 'IMS AS MediaControl Service'
  version: 1.0.0-alpha.2
  description: |
    Nimsas_MediaControl Service.
    © 2024, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V18.0.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175'

servers:
  - url: '{apiRoot}/nimsas-mc/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-mc

paths:

  /call-sessions/{sessionId}/media-instruction:
    post:
      summary: Update
      operationId: UpdateCallSession
      tags:
        - Individual call session (Document)
      parameters:
        - name: sessionId
          in: path
          description: Session ID assigned by the IMS AS during the IMS session setup.
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
      requestBody:
        description: Representation of the updates to apply to the call session.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/MediaInstructionData'
      responses:
        '200':
          description: >
            Successful update of the call session, when the IMS AS needs to return information
            in the response.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/MediaInstructionData'
        '204':
          description: >
            Successful update of the call session, when the IMS AS does not need to return
            information in the response.
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target UPF.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
```

```
            Location:
              description: The URI pointing to the resource located on the redirect target UPF.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '501':
          $ref: 'TS29571_CommonData.yaml#/components/responses/501'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-mc: Access to the Nimsas_MediaControl API.

  schemas:

# STRUCTURED DATA TYPES

    MediaInstructionData:
      description: The media instruction data for a specific session.
      type: object
      required:
        - sessionId
        - mediaInstructionSet
      properties:
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        mediaInstructionSet:
          description: >
            Contains a set of instructions for each media flow to control. The mediaId
            attribute within the MediaInstructions data type is the key of the map.
          type: object
          additionalProperties:
            $ref: '#/components/schemas/MediaInstructions'
          minProperties: 1

    MediaInstructions:
      description: The media instructions for a specific media flow.
      type: object
      required:
        - mediaId
        - mediaResourceType
      properties:
        mediaId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaId'
        mediaResourceType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaResourceType'
        mediaInstruction:
          $ref: '#/components/schemas/MediaInstruction'
        dcMediaSpecification:
```

```
          $ref: '#/components/schemas/DcMediaSpecification'
        arMediaSpecification:
          $ref: '#/components/schemas/ArMediaSpecification'
        mediaProcessingUrl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'

    DcMediaSpecification:
      description: The data channel media specification information.
      type: object
      required:
        - streams
      properties:
        mediaProxyConfig:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaProxy'
        replaceHttpUrls:
          description: >
            Contains a list of replacement HTTP URLs. The streamId
            attribute within the ReplaceHttpUrl data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ReplaceHttpUrl'
          minProperties: 1
        mdc1EndpointDcsf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        mdc1EndpointMf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        mdc2EndpointDcAs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        mdc2EndpointMf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        mdc2Protocol:
          $ref: 'TS29176_Nmf_MRM.yaml#/components/schemas/Mdc2Protocol'
        streams:
          description: >
            Contains a data channel mapping and configuration information. The streamId
            attribute within the DcStream data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/DcStream'
          minProperties: 1

    ArMediaSpecification:
      description: The AR media specification information.
      type: object
      required:
        - mediaProcessingSpec
      properties:
        mediaProcessingSpec:
          type: string

# ENUMS

    MediaInstruction:
      description: The media instruction for handling a specific media flow.
      anyOf:
        - type: string
          enum:
            - TERMINATE_MEDIA
            - ORIGINATE_MEDIA
            - TERMINATE_AND_ORIGINATE_MEDIA
            - UPDATE_MEDIA
            - DELETE_MEDIA
            - REJECT_MEDIA
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.
```

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2023-04	CT4#115e	C4-231514				TS skeleton	0.0.0
2023-04	CT4#115e	C4-231551				Inclusion of pCRs agreed at CT4#115e, including C4-231214, C4-231217, C4-231224, C4-231225, C4-231514, C4-231515, C4-231516, C4-231517.	0.1.0
2023-05	CT4#116	C4-232553				Inclusion of pCRs agreed at CT4#116, including C4-232142, C4-232145, C4-232504, C4-232505.	0.2.0
2023-09	CT4#117	C4-233810				Inclusion of pCRs agreed at CT4#117, including C4-233250,C4-233254,C4-233255,C4-233256,C4-233700,C4-233701,C4-233704,C4-233856.	0.3.0
2023-10	CT4#118	C4-234583				Inclusion of pCRs agreed at CT4#118, including C4-234172, C4-234174, C4-234175, C4-234177, C4-234545, C4-234547.	0.4.0
2023-12	CT4#119	C4-235660				Inclusion of pCRs agreed at CT4#119, including C4-235221, C4-235464, C4-235465, C4-235466, C4-235467, C4-235468, C4-235471	0.5.0
2023-12	CT#102	CP-233022				TS presented for information	1.0.0
2024-03	CT4#121	C4-240852				Inclusion of pCRs agreed at CT4#121, including C4-240405, C4-240406, C4-240407, C4-240409, C4-240410, C4-240470, C4-240704, C4-240705, C4-240708, C4-240716	1.1.0
2024-03	CT#103	CP-240025				TS presented for approval	2.0.0
2024-03	CT#103					TS approved in CT#103	18.0.0

History

Document history		
V18.0.0	May 2024	Publication